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Collision Risk Monitoring System

1. Introduction

This project is a Collision Risk Monitoring System designed to simulate and display how nearby targets around a ship are evaluated for collision risk.
It uses a Python backend to generate movement data and a web interface to visualize the results.

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2. Problem Understanding

In real marine navigation, ships need to continuously monitor nearby vessels.
The most important factor is how close another ship will come in the future, which is called **CPA (Closest Point of Approach)**.

If CPA is very small, the collision risk is high.
So the system must:

- Track targets
- Calculate CPA
- Show risk level

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3. Solution Design

The solution is divided into two parts:

Backend (Python)

- Simulates ship and target movement
- Calculates CPA values
- Writes the data into a JSON file

Frontend (HTML + JavaScript)

- Reads the JSON file
- Draws targets on a canvas
- Displays risk alerts

This separation makes the system simple and scalable.

4. CPA Calculation

CPA is calculated using basic distance and motion formulas.
It estimates the shortest distance that will occur between the ship and a target if both continue moving.

This value is then compared with fixed limits to decide the risk level.

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5. Alert System

Based on CPA:

- **Safe** → Green
- **Warning** → Orange
- **Danger** → Red

These colors are shown in the alert panel to quickly understand the situation.

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6. Testing and Verification

The system was tested by:

- Running the Python simulation
- Checking whether JSON data is updated
- Verifying that the web page shows the same values
- Confirming that colors and CPA values change correctly

The output was consistent with the calculations, so the system is working correctly.

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7. Conclusion

This project successfully demonstrates how collision risk can be simulated and visualized using simple tools.
It clearly shows how data from a Python backend can be used in a real-time web interface to support safety decisions.